

Inventory Intelligence Demand-Driven Reorder Point

A data-driven framework for balancing stock levels and reorder timing —
built on 1 million real retail transactions.



The Cost of Getting It Wrong

Too Much Inventory

Excess holding costs, tied-up capital, obsolescence risk

Too Little Inventory

Stockouts, lost sales, damaged customer trust

The Goal

Find the precise reorder point that minimizes both





DATASET

UCI Online Retail II

1 Million
Transactions

Across 2 years
(2009–2011)

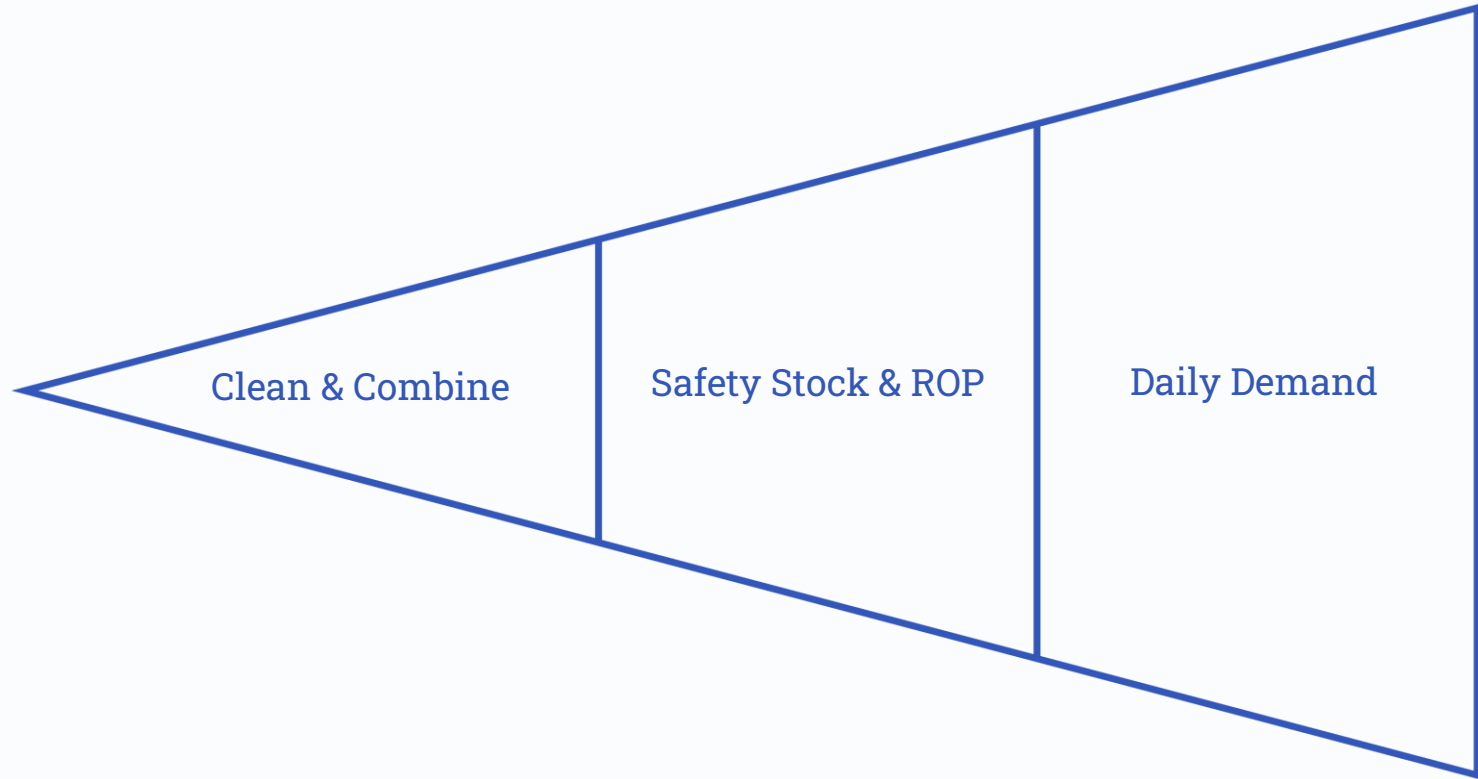
UK-Based
Online Retail

Product codes,
quantities, prices,
dates

Real
Transaction
Data

No synthetic or
simulated source
data

From Raw Data to Reorder Points



Each step transforms raw transaction logs into actionable inventory intelligence — from cleaning to dashboard.

The Core Inventory Formulas

Safety Stock

$$Z \times \sigma \times \sqrt{\text{Lead Time}}$$

Buffers against demand variability

Reorder Point (ROP)

$$\text{Avg Demand} \times \text{Lead Time} + \text{Safety Stock}$$

Triggers the reorder signal

Service Level

$$95\% \rightarrow Z = 1.65$$

Industry-standard target

$$\#476FD = (1 \times (\%2(\%))$$

$$A \quad X_1 = (x_0(\% + \square + \frac{9}{10}(\%))$$

$$D \quad \square^2(x) = x_1^a = (1^6 + 0)$$

$$B \quad \square \nabla a^a = (0^x / + - \hat{A}^x + 9)$$

$$N_1 = \frac{1}{2} + \frac{2}{8} + (B \neq \frac{100}{100})$$

$$E \quad 1/3 p^0 \square H(B + 8^9) + \frac{109}{100}$$

$$\#7521 = 0(\frac{2}{100})$$

ASSUMPTIONS

Documented & Transparent

Where real data wasn't available, we used industry-standard placeholders – all clearly documented.



→ Lead Time: 7 days
Industry-standard placeholder

→ Current Stock: Simulated
No live inventory in source data

→ Service Level: 95%
Z-score = 1.65

→ Unit Price: Estimated
 $\text{Revenue} \div \text{quantity sold}$

ABC ANALYSIS

A Small % Drives Most Revenue

Category A

~20% of SKUs → ~80% of revenue

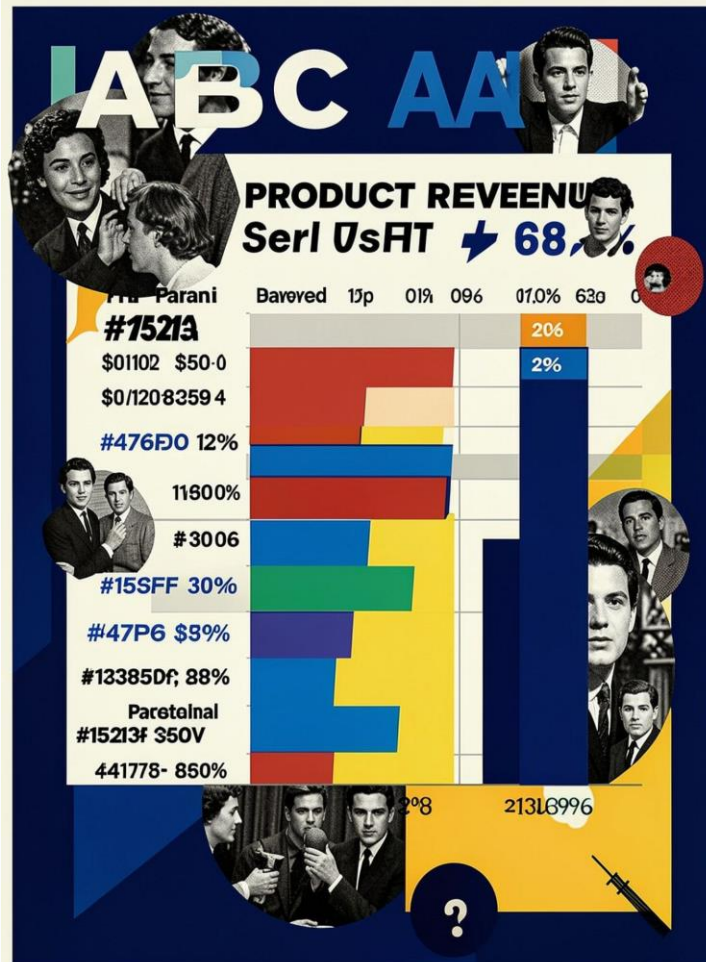
Tightest reorder accuracy required

Category B

Moderate revenue contributors
Standard monitoring cadence

Category C

Long tail of low-revenue SKUs
Simplified reorder rules



Interactive Inventory Intelligence



KPI Cards

Total SKUs, reorder alerts, estimated inventory value



Reorder Alert Table

ABC-prioritized list of SKUs needing restock



ABC Revenue Breakdown

Visual share of revenue by category



Interactive Filters

Drill down by category and reorder status



TECH STACK

Tools Behind the Analysis



Python

Pandas + NumPy for data cleaning, transformation, and formula calculations



Power BI

DAX measures, data modeling, and interactive visualizations



Excel

Source data ingestion and intermediate data validation

Key Takeaway

Category A products demand the tightest reorder accuracy — stockouts here are the most costly to the business.

Data-Driven

Real transactions, not estimates

Actionable

Clear reorder signals per SKU

Scalable

Framework reusable across catalogs

